WORKED SOLUTION

STEP I 1993, Question 9

Pure • First-Order Differential Equations

STEP I 1993, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q9

The key idea

This is a mixing problem: the rate of change of the amount of A in a vat is (rate in) minus (rate out), and because mixing is rapid the liquid leaving has the same concentration as the vat. The first thing to notice is that each vat's total volume never changes, so the concentrations are just "amount of A divided by a constant". Each vat then gives a first-order linear equation, and the output of the first vat becomes the forcing term for the second.

Solution

The first vat

Liquid enters the first vat at a total rate $a + (b - a) = b$ litres per second and leaves at b litres per second, so its volume stays at its initial value K litres for all $t \geq 0$.

Mixing is rapid, so at time t the contents are uniform, and the fraction of the liquid that is chemical A is $V(t)/K$. Over a short time δt, $a\,\delta t$ litres of A flow in, and $b\,\delta t$ litres of mixture flow out, of which $\frac{V}{K}\,b\,\delta t$ litres are A. Hence

$$\delta V \approx a\,\delta t - \frac{bV}{K}\,\delta t,$$

and dividing by δt and letting $\delta t \to 0$,

$$\dot{V}(t) = -\frac{bV(t)}{K} + a,$$

which is the required equation.

The equation is linear with constant coefficients. Writing it as $\dot{V} + \frac{b}{K}V = a$ and multiplying by the integrating factor $\mathrm{e}^{bt/K}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(V\mathrm{e}^{bt/K}\right) = a\,\mathrm{e}^{bt/K} \quad \Longrightarrow \quad V\mathrm{e}^{bt/K} = \frac{aK}{b}\mathrm{e}^{bt/K} + C.$$

Initially the vat holds only chemical B, so $V(0) = 0$, giving $C = -\frac{aK}{b}$. Therefore

$$V(t) = \frac{aK}{b}(1 - \mathrm{e}^{-bt/K}), \qquad t \geq 0,$$

as required.

The second vat

The second vat receives b litres per second from the first and empties at b litres per second, so its volume stays at L litres. Let $W(t)$ be the volume of A in it at time t, so $W(0) = 0$.

Ask yourself At what rate does chemical A arrive in the second vat?

It arrives at exactly the rate it leaves the first vat, namely $\frac{bV}{K} = a(1 - \mathrm{e}^{-bt/K})$. It leaves at rate $\frac{bW}{L}$, by the same rapid-mixing argument. So

$$\dot{W} + \frac{b}{L}W = a - a\,\mathrm{e}^{-bt/K}.$$

Multiplying by the integrating factor $\mathrm{e}^{bt/L}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(W\mathrm{e}^{bt/L}\right) = a\,\mathrm{e}^{bt/L} - a\,\mathrm{e}^{bt(1/L - 1/K)}.$$

Because $L \neq K$, the exponent $b(\frac{1}{L} - \frac{1}{K}) = \frac{b(K-L)}{KL}$ is non-zero, so we can integrate the second term as an exponential:

$$W\mathrm{e}^{bt/L} = \frac{aL}{b}\mathrm{e}^{bt/L} - \frac{aKL}{b(K-L)}\mathrm{e}^{b(K-L)t/(KL)} + D.$$

Multiplying through by $\mathrm{e}^{-bt/L}$, and noting $\frac{b(K-L)}{KL} - \frac{b}{L} = -\frac{b}{K}$,

$$W = \frac{aL}{b} - \frac{aKL}{b(K-L)}\mathrm{e}^{-bt/K} + D\mathrm{e}^{-bt/L}.$$

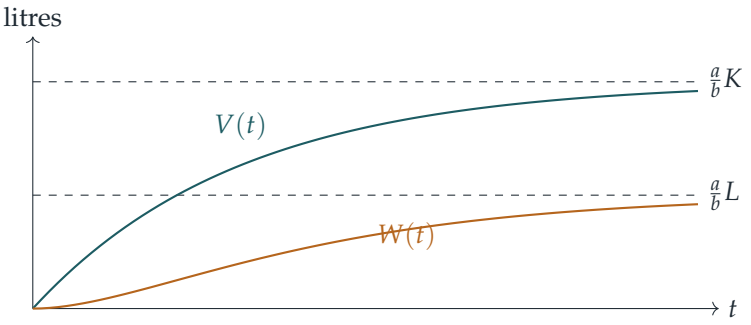
Setting $W(0) = 0$:

$$D = \frac{aKL}{b(K-L)} - \frac{aL}{b} = \frac{aL}{b} \cdot \frac{K - (K-L)}{K-L} = \frac{aL^2}{b(K-L)}.$$

So the second vat contains

$$\boxed{W(t) = \frac{aL}{b}\left(1 - \frac{K\mathrm{e}^{-bt/K} - L\mathrm{e}^{-bt/L}}{K-L}\right)}$$

litres of chemical A at time $t \geq 0$.



The sketch uses $a = 1$, $b = 2$, $K = 10$, $L = 5$. Notice that W starts with zero gradient: at first there is almost no A in the liquid arriving from the first vat.

Checks

- At $t = 0$ the bracket is $1 - \frac{K-L}{K-L} = 0$, as it should be. Also $\dot{W}(0) = 0$, matching the fact that A arrives at rate $bV(0)/K = 0$.
- As $t \to \infty$, $V \to \frac{a}{b}K$ and $W \to \frac{a}{b}L$: both vats end up with concentration a/b of A, which is the concentration of the incoming liquid.
- The formula is symmetric in K and L apart from the factor L in front, which is a nice consistency check on the algebra.
- With $a = 1$, $b = 2$, $K = 10$, $L = 5$, $t = 5$: $V = 5(1 - \mathrm{e}^{-1}) \approx 3.1606$ and $W = 2.5(1 - \mathrm{e}^{-1})^2 \approx 0.99894$. A numerical simulation of both vats (tracking volumes too) agrees with both formulas to about 10^{-13} for many random choices of a, b, K, L.

Common slips

- Using $b - a$ or a for the outflow of A: the outflow is the total outflow rate times the concentration.
- Taking the inflow of A to the second vat to be a: only the A that leaves the first vat reaches the second.
- Forgetting why $L \neq K$ is needed: if $L = K$ the forcing term $\mathrm{e}^{-bt/K}$ solves the homogeneous equation, and the answer instead contains a $t\,\mathrm{e}^{-bt/K}$ term.

Beyond STEP

A chain of well-mixed tanks is a standard compartment model, used for drug doses moving through the body and for radioactive decay chains. Written as a system $\dot{\mathbf{x}} = M\mathbf{x} + \mathbf{f}$, the decay rates b/K and b/L are the eigenvalues of M, and the case $L = K$ is a repeated eigenvalue, the same "resonance" that gives $t\,\mathrm{e}^{\lambda t}$ solutions. Boyce

and DiPrima's *Elementary Differential Equations* treats both mixing problems and linear systems in this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.